

SLIP – 17

Section I: Data Structures [15 Marks]

Q 1) Perform the following operations using implementation of stack (Static or dynamic)

- i. Accept n integers from user and push it into the stack. ii. Reverses a string of characters.**

Answer:

```
#include <stdio.h>
#include <string.h>
#define MAX 100

int stack[MAX], top = -1;
char cstack[MAX]; int ctop
= -1;

void push(int x)
{
    if(top == MAX - 1)
        printf("Stack Overflow\n");
    else
        stack[++top] = x;
}

int main()
{
    int n, x, i;
    char str[MAX];

    printf("Enter n: ");
    scanf("%d", &n);

    printf("Enter %d integers:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &x);
        push(x);
    }

    printf("Stack elements: ");
    for(i = top; i >= 0; i--)
        printf("%d ", stack[i]);

    printf("\nEnter a string: ");
    scanf("%s", str);
```

```

    for(i = 0; str[i] != '\0'; i++)
        cstack[++ctop] = str[i];

    printf("Reversed string: ");
    while(ctop != -1)        printf("%c",
cstack[ctop--]);    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database (Primary keys are underlined)

Property (Pno, area_Sqft, location, city)

Owner (oname, address, phone)

The relationship is as follows: Property – Owner are related with Many to One relationship.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Owner(
    oname VARCHAR(50) PRIMARY KEY,
    address VARCHAR(100),    phone
    VARCHAR(15)
);

CREATE TABLE Property(    Pno INT PRIMARY KEY,
    area_Sqft INT,    location VARCHAR(50),
    city VARCHAR(50),    oname VARCHAR(50)
REFERENCES Owner(oname)
);

INSERT INTO Owner VALUES
('Mr.Rahane', 'Pune', '9822000001'),
('Mr.Baravkar', 'Pune', '9822000002'),
('Mr.Joshi', 'Nashik', '9822000003');

INSERT INTO Property VALUES
(1,2500, 'Kothrud', 'Pune', 'Mr.Rahane'),
(2,12000, 'Kothrud', 'Pune', 'Mr.Baravkar'),

```

```
(3,1500,'College Road','Nashik','Mr.Joshi'),  
(4,20000,'Kothrud','Pune','Mr.Joshi');
```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks].

Answer:

```
SELECT COUNT(*) FROM Property WHERE  
oname = 'Mr.Rahane';
```

i. Count all properties of “Pune” city.

Answer:

```
SELECT COUNT(*) FROM Property  
WHERE city = 'Pune';
```

ii. Delete all properties in “Pune” city owned by “Mr.Baravkar”.

Answer:

```
DELETE FROM Property  
WHERE city = 'Pune'  
AND oname = 'Mr.Baravkar';
```

Q 2(C)

C. Write a query to find the names of all owners who own a property in Pune city, located in ‘Kothrud’ area and which has an area greater than 10000sqft.

Answer:

```
SELECT DISTINCT oname  
FROM Property  
WHERE city = 'Pune'  
AND location = 'Kothrud'  
AND area_Sqft > 10000;
```

Q 3) Viva [5 marks]